COMP 7810A Stablecoins, Tokenization, and RWA
Assignment Two — Stablecoin Lab
Due Date: 16 October 2026 23:55pm

*** Submit one zip of the project (keep the lib/ directory inside) to Moodle ***

1) [40%]

Contract correctness and test coverage. Work through Ex0–Ex6 in `EXERCISES.md`; `make exercise` must end up all green.

(i) Ex0 — `make doctor`, `make test` all green.
(ii) Ex2 — the two `test_Ex2_*` tests in `01_LoopTasks.t.sol`.
(iii) Ex4 — the five `test_Ex4_*` tests in the same file.
(iv) Ex5 — the four TODOs in `OverCollateralizedVault.sol`.
(v) Ex6 — the handler's `redeem` plus two `invariant_*` tests.

Ex1 is a warm-up and is **not graded**. Ex3 is also run by hand — its screenshot of `totalSupply()` far exceeding `totalCollateral()` is required. A local Anvil is enough.

2) [20%]

Testnet deployment and verification (Tier 2). Deploy to Sepolia, verify on Etherscan, and paste the three addresses and links into the **Moodle text box**. Use a throwaway wallet; the key stays in `.env`.

3) [25%]

Threat-model analysis. Answer groups A–D in `STUDENT-QUESTIONS.md`, 150–300 words each, plus the section-E table (the five Ex4 behaviours and the one scenario most likely to be attacked).

4) [15%]

Code style and documentation. Clear naming, comments that explain *why*, and an architecture diagram in `README.md` (a photo of a hand drawing is fine).

What to submit

Build it from the project root. The zip must contain the Ex3 screenshot, your answers, and the architecture diagram. **A private key anywhere in the zip scores zero.**

Optional (Tier 3, unmarked)

Ex7 (`make challenge`), a Peg Stability Module, a tighter burn permission, or real Chainlink instead of `MockPriceFeed`.

--- End of Assignment ---